More Patients, Fewer Bleeds?

Anticoagulation Safety in England 2005–2025 — Executive Summary

stepchangeanalysis.com | Bootstrap CUSUM SPC Analysis | May 2026

Note: All rates are surrogate indicators of structural change derived from MHRA Yellow Card reports and NHS BSA Prescription Cost Analysis data. The structural change points are robust; absolute rate figures are directional estimates.

Five Confirmed Findings

1

Anticoagulant patient population structurally doubled

Bootstrap CUSUM detects a structural explosion in total BNF 0208 prescriptions — from a Stage 1 mean of 8.8M to 18.3M items/year. Any analysis of raw adverse event counts without correcting for this denominator change is invalid. Confirmed at 95% confidence (98.5%).

Change point: 2015 | Effect: +107%

2

Total adverse event rate stepped up as DOACs arrived

Rate per million prescriptions rose from Stage 1 mean of 35/M to Stage 2 mean of 79/M. Within Stage 2, 2015–2016 peaks of ~137/M breach the X-mR upper natural process limit (UNPL=97.15) — genuine special cause signals confirmed by two independent methods.

Change point: 2012 | Effect: Step up

3

DOAC adverse event rate structurally fell by over two thirds

DOAC rate stepped down from ~650/M to ~185/M at 95% confidence. Driven by ROCKET-AF trial controversy (Nov 2015), EMA rivaroxaban renal dosing label update, and rapid prescriber switch from rivaroxaban to apixaban. Confirmed by COBRRA trial (NEJM, March 2026): apixaban 3.3% bleeding vs rivaroxaban 7.1%.

Change point: 2016 | Effect: -72%

4

Warfarin population became higher risk as lower-risk patients switched to DOACs

As lower-risk patients migrated to DOACs, the remaining warfarin population became progressively smaller and sicker. Warfarin rate rose from ~13/M in 2011 to ~40/M in 2020. Marginal structural change at 90% confidence only.

Change point: 2014 | Effect: 90% only

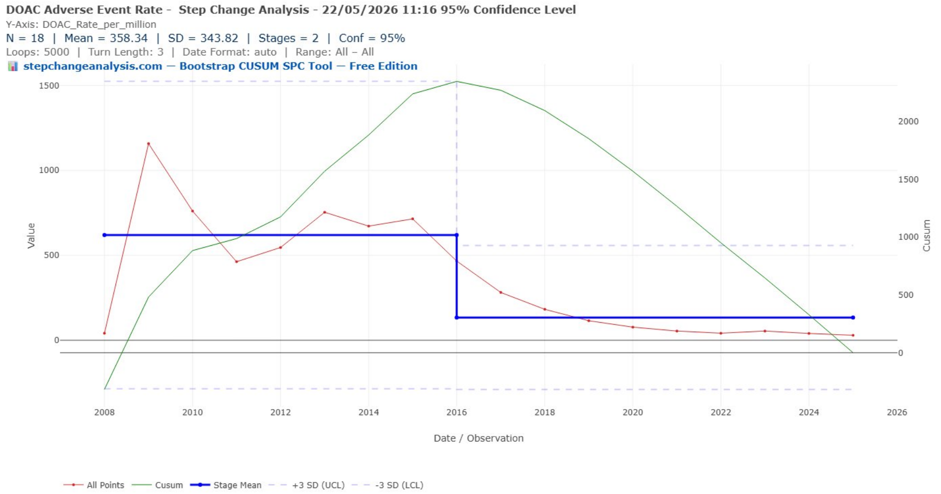
5

Prescribing quality problems remain despite national improvement

Even by 2019, ~50% of DOACs were still not adjusted for renal function (Burn et al., 2023). Inappropriate prescribing in 8.4–28.9% of hospitalised patients (Malhotra et al., 2023). Regional variation 53–99% (Bassi et al., 2020). 9–20% of DOAC patients on off-label doses (Dawson et al., 2020; Shen et al., 2021).

Change point: Now | Effect: Still high

Key Chart: DOAC Adverse Event Rate — Structural Step Down at 2016



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DOAC adverse event rate per million DOAC prescriptions, England 2008–2025. Bootstrap CUSUM 95% confidence detects structural step down at 2016. The CUSUM found the signal; the NEJM confirmed the reason a decade later (COBRRA trial, March 2026).

The Key Argument: The Job Is Not Done

The national DOAC adverse event rate fell 96% from peak. But the national average masks substantial and persistent local variation. Even by 2019, approximately half of all DOAC prescriptions were still not adjusted for renal function. Inappropriate prescribing occurs in up to 29% of hospitalised patients. Studies show 9–20% of DOAC patients are on an off-label dose. Anticoagulants have now overtaken antibiotics and prescription opioids as the number one cause of adverse drug events (Budnitz et al., JAMA, 2021). Across English regions, DOAC prescribing rates vary from 53% to 99% — variation not explained by patient population differences.

The national average has improved. The local variation — and the scope for further improvement — remains large.

A nationwide VA study (Barnes et al., J Am Heart Assoc, 2024) across 128,652 patients at 123 centres found that early adoption of a DOAC population management dashboard was associated with reduced off-label dosing, reduced bleeding, and all sites experienced reductions in VTE and stroke following adoption. Bootstrap CUSUM applied to local adverse event rate data is precisely the tool to detect whether such improvements are genuine structural changes or common cause noise.

Analytical Framework: Deming & Joiner

Concept	Applied to anticoagulation
Common cause vs special cause (Deming, Out of the Crisis, 1982)	2015–2016 ADR peaks breach X-mR UNPL — genuine special cause. 2016 DOAC step down confirmed at 95% — driven by identifiable external events.
Tampering (Joiner, Fourth Generation Management, Ch.8, p.128)	Adjusting warfarin doses in response to normal INR variation destabilises the process. The funnel experiment applied to anticoagulation clinics.
Stratify (Joiner)	National average looks acceptable. Stratify by drug, renal function, or practice — the variation is substantial and actionable.
Experiment (Joiner)	Evaluate prescribing improvement programmes with Bootstrap CUSUM before and after. A genuine structural improvement produces a new stage boundary with high confidence.

Key References

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• Full article and interactive data tool: stepchangeanalysis.com/anticoag-safety.html